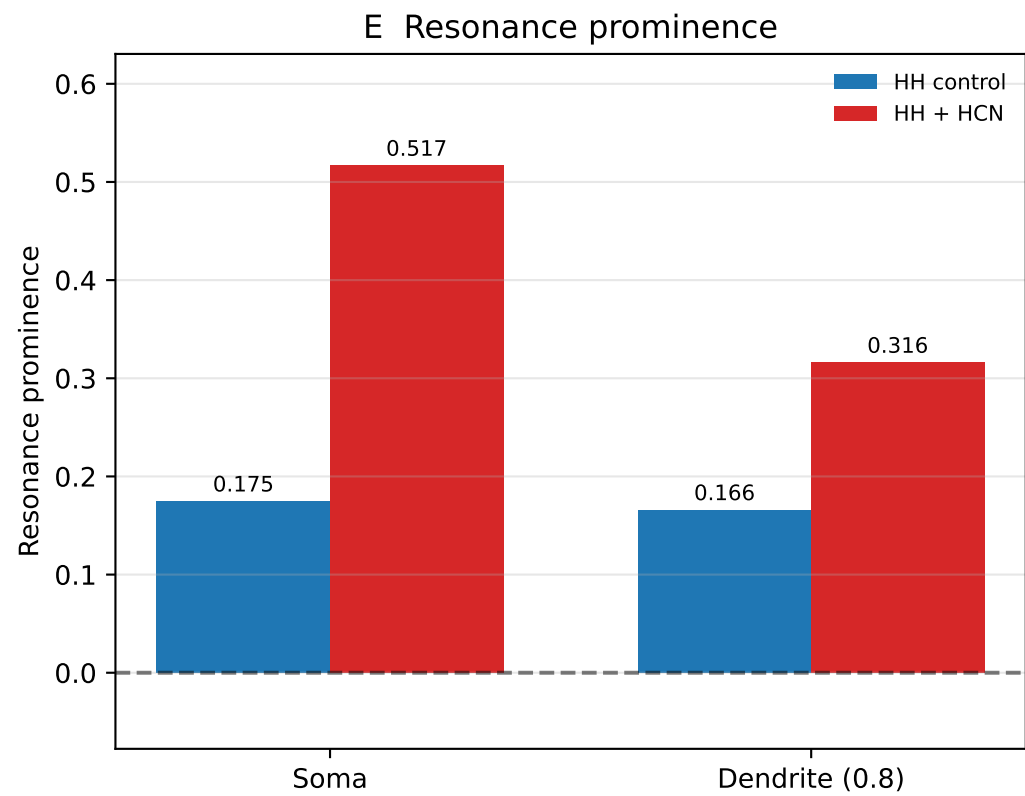
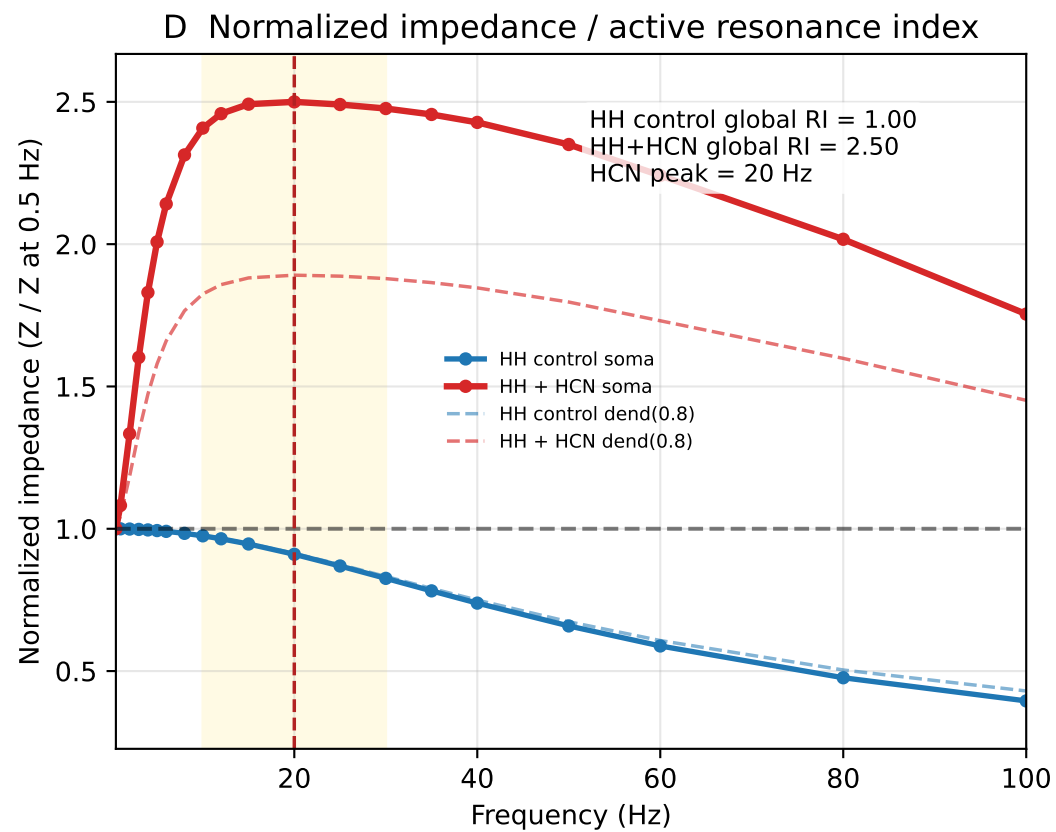
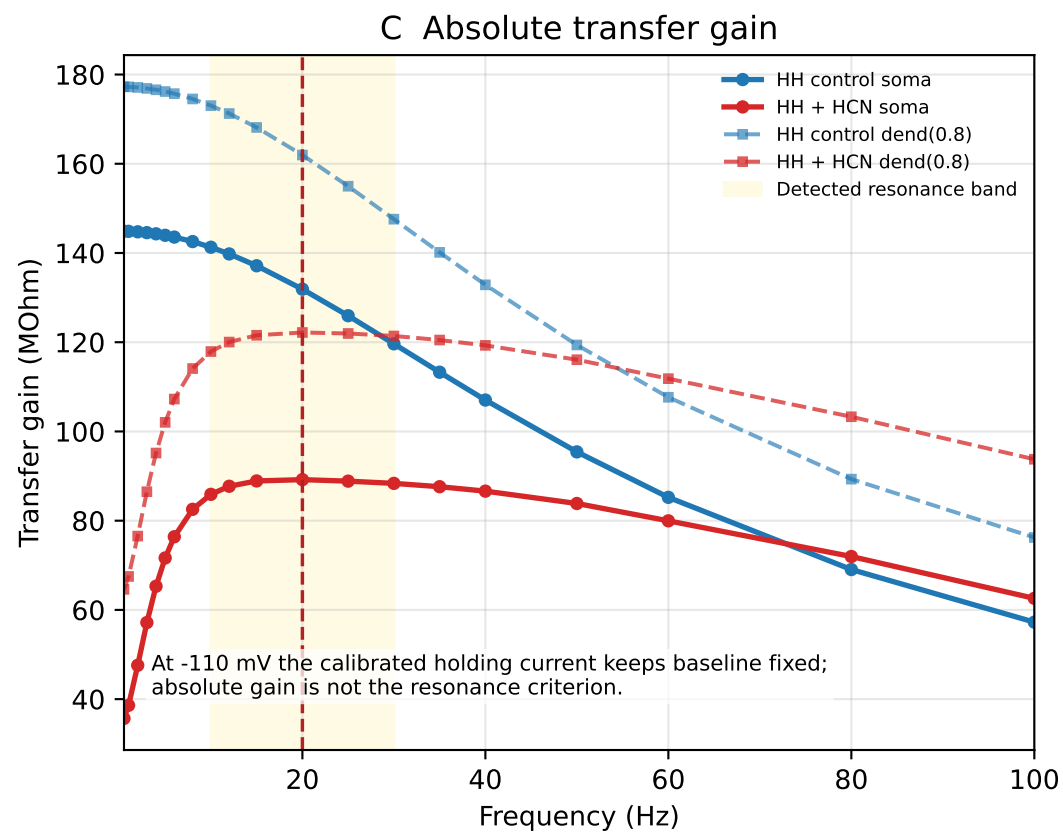
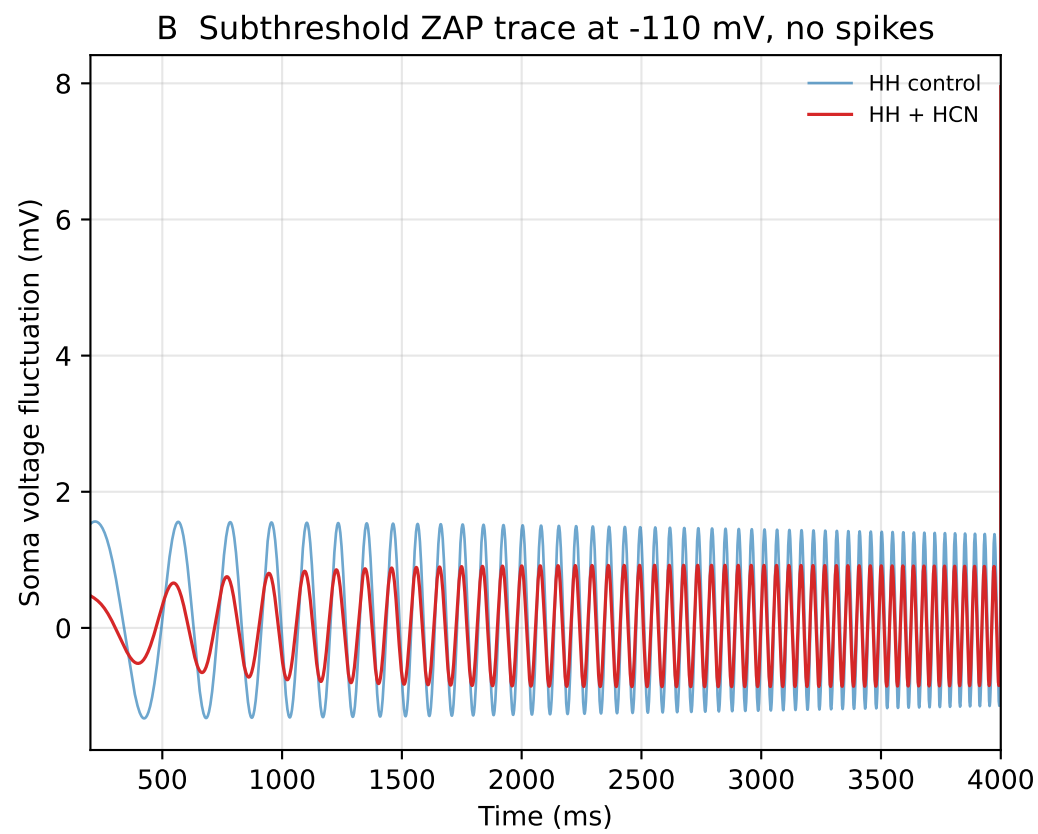
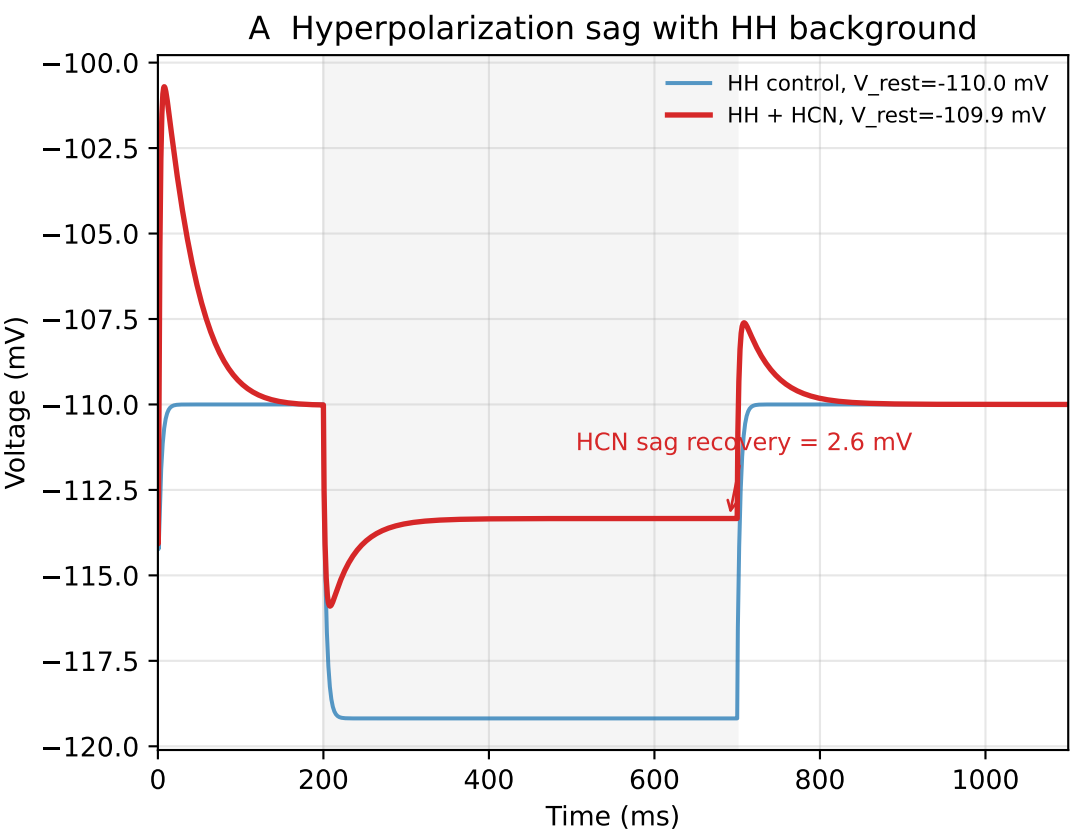


HH + HCN/Ih model: -110 mV subthreshold active resonance



Model:

- Soma: pas + hh1 Na/K +/- Ih
- Dendrite: pas +/- Ih
- Input: distal current at dend(0.8)
- Recording: soma and dend(0.8)

Operating point:

- Operating Vm: -110.0 mV
- Holding control: -0.2450 nA -> soma -110.0 mV, dend -100.5 mV
- Holding HCN: -0.4027 nA -> soma -110.0 mV, dend -95.2 mV
- Holding: calibrated current clamp, not voltage clamp
- Fixed g_h : 0.0008 S/cm²
- Soma Ih density: 0.25 x g_h
- AC amplitude: 0.0100 nA
- Frequency range: 0.5-100 Hz
- Resonance band: 10.0-30.0 Hz
- HH effects are expected to be weak at this hyperpolarized Vm.

Validation:

- HH spike validation: True, current=0.08 nA
- ZAP/sweep for resonance: no spikes = True

Results:

- HCN sag recovery: 2.56 mV
- HH control global RI: 1.000
- HH+HCN global RI: 2.500
- HH+HCN global peak: 20.0 Hz
- Theta-band RI: 2.408
- Prominence soma: control=0.175, HCN=0.517
- Prominence dend: control=0.166, HCN=0.316

Conclusion:

HH+HCN global RI > HH control global RI under -110 mV subthreshold conditions. This reflects normalized frequency tuning, not absolute voltage amplification.